CRYSTALLOGRAPHY & X-RAY DIFFRACTION — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Crystallography & X-ray Diffraction

Question 1

10.0 points

Analyze the diagram below illustrating the geometry of Bragg's law, where an incident X-ray beam strikes a crystal plane, and the diffracted beam is observed. Identify the angle of incidence, the spacing between planes, and the conditions for constructive interference according to the equation $n\lambda = 2d\sin\theta$. Explain how each component is represented in the diagram.

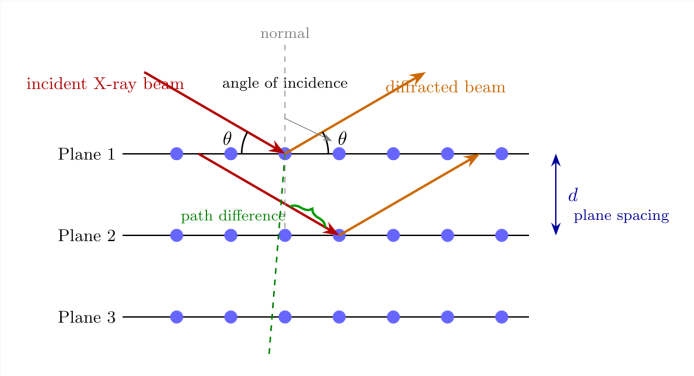

Fig. 1. Diagram illustrating the geometry of Bragg's law. Show an incident X-ray beam striking a crystal plane at an angle θ, with the diffracted beam also at angle θ. Include multiple parallel crystal planes with spacing d between them. Label the incident angle θ, the plane spacing d, and the path difference between the beams. Indicate the condition for constructive interference using the equation nλ = 2dsinθ. IMPORTANT correction needed: The diagram explicitly shows the Bragg's Law equation. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 2 10.0 points

Examine the diagram below of the reciprocal lattice and Ewald sphere construction. Determine where the reflections will occur when an X-ray beam is incident on a crystal. Define the conditions under which a specific point in the reciprocal lattice satisfies the Laue condition, and interpret its meaning in the context of X-ray diffraction.

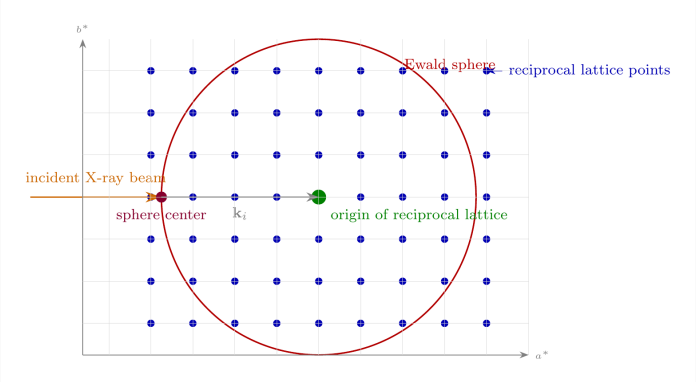

Fig. 2. Draw a diagram showing the reciprocal lattice and Ewald sphere construction. Include the incident X-ray beam, the Ewald sphere centered at the origin of the reciprocal lattice, and several reciprocal lattice points. Highlight the points where the Ewald sphere intersects the reciprocal lattice, indicating potential reflection conditions. Label the incident beam, the Ewald sphere, and the reciprocal lattice points. IMPORTANT: The diagram is missing required labels. The diagram is missing a label for the "origin of. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing a. Add each missing label as visible text at the appropriate location in the diagram.

Question 3 10.0 points

Interpret the diagram below showing Miller indices (100), (110), and (111) planes in a cubic unit cell. Explain how these planes are oriented with respect to the cubic axes and how each set of indices determines the plane's orientation.

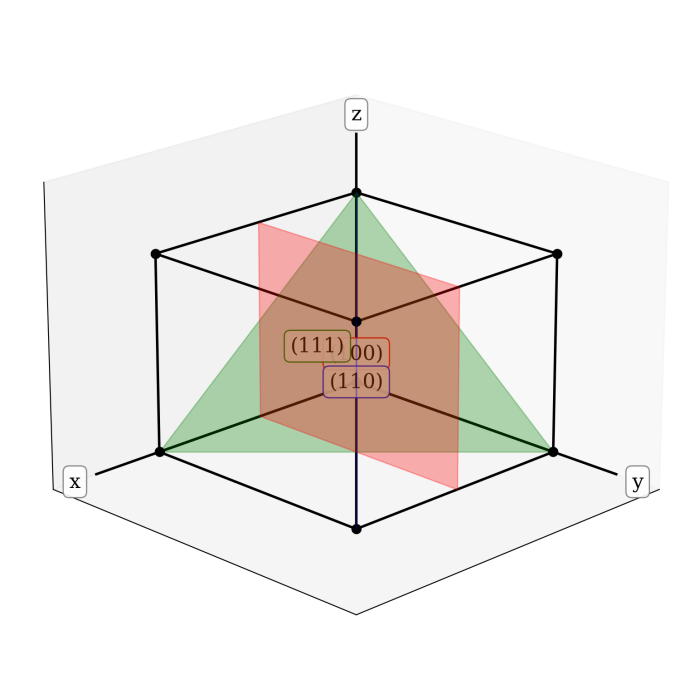

Fig. 3. 3D diagram of a cubic unit cell showing the Miller indices (100), (110), and (111) planes. The unit cell should be represented as a cube with axes labeled x, y, and z. The (100) plane should be shown parallel to the yz-plane, the (110) plane should be diagonal in the xy-plane, and the (111) plane should cut through all three axes equally. Each plane should be labeled with its respective Miller index. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly labels the planes; the plane. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly labels the planes; the plane labeled. Regenerate using EXACTLY the values and sequences specified.

Question 4

10.0 points

Examine the diagram below of an FCC diffraction pattern. Identify the systematic absences in the diffraction peaks and explain the underlying cause of these absences based on the lattice plane reflections.

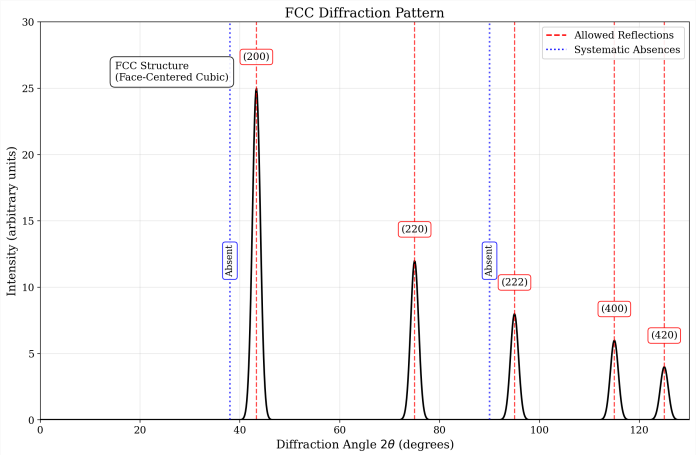

Fig. 4. CRITICAL: The previous diagram showed the wrong type. The diagram description explicitly requests a 1D diffraction. Regenerate from scratch showing the CORRECT diagram type. An FCC diffraction pattern showing diffraction peaks. The diagram should include labeled axes for the diffraction angle and intensity, with peaks corresponding to the allowed reflections of an FCC lattice. Systematic absences should be indicated as missing peaks where reflections are expected based on the lattice structure. IMPORTANT correction needed: The diagram explicitly labels the "Systematic Absences". Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Examine the atomic arrangement of the BCC crystal in the diagram below. Use this to calculate the structure factor for a given set of Miller indices, assuming only atomic scattering factors are considered. Detail the steps involved in the calculation.

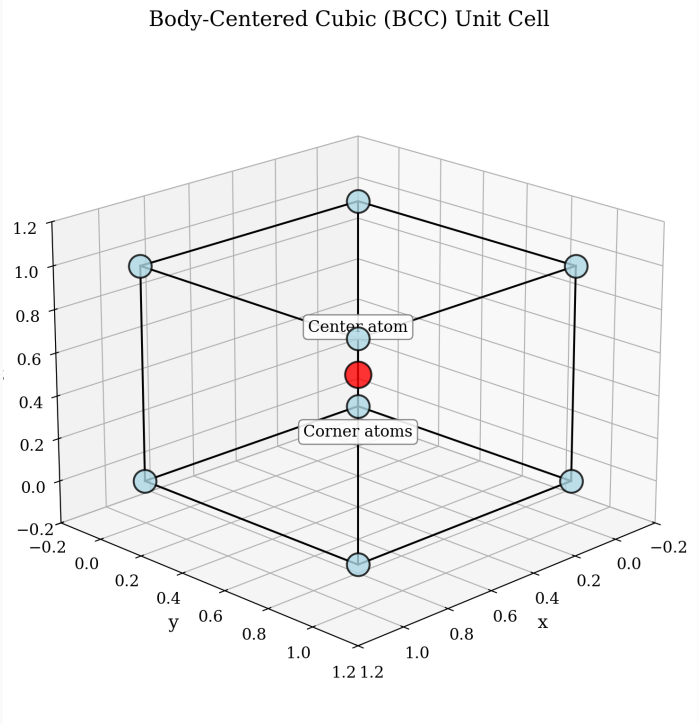

Fig. 5. 3D diagram of a BCC (Body-Centered Cubic) unit cell. The diagram should show a cube with atoms at each corner and a single atom at the center of the cube. Each atom should be represented as a sphere, and the positions should be labeled to indicate the body-centered arrangement. The diagram should highlight the spatial arrangement of atoms within the unit cell, which is essential for calculating the structure factor for given Miller indices.